Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/08/2007
PI Name:	Lupton, Joanne R. Ph.D.		
Project Title:	Nutritional Countermeasures to Radiation-Enhanced Colon Cancer		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Chapkin, Robert (Texas A&M University) Ford, John (Texas A&M University) Turner, Nancy (Texas A&M University) Carroll, Raymond (Texas A&M University)		
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Task Description:	1. Original Aims. The primary goal of this project is to develop a set of colon cell gene expression profiles that accurately characterize the radiation-enhancing effect on colon tumor development and the chemoprotective effect of dietary countermeasures in this process. This goal is addressed by specific aim #1: From the global transcriptional profile (analyzed by microarray technology) in samples isolated from colonic mucosa, determine which sets of gene expression profiles best characterize the initiation stage, the promotion stage (aberrant crypt development) and the final tumor stage of colon cancer development using a 2 x 4 x 4 experimental design (+ or – radiation; four time points and 4 diets). The secondary goal, addressed by specific aim #2, is to compare gene expression profiles developed from exfoliated cells to those from colonic mucosal cells to determine how well they reflect in vivo events. This information is important for our short-term future goal of using fecal material from humans as a noninvasive method to monitor changes in gene expression patterns over time. Our long-term future goal is to have sets of gene expression profiles that can be used to monitor astronauts before, during, and after space flight to detect exposure and response to a radiation event so that appropriate intervention strategies (e.g., dietary countermeasures) can be implemented as needed. Data generated from this research program will support the NASA/NSBRI goals of: 1) gaining greater understanding of the mechanisms involved in the synergy of radiation and other environmental insults on carcinogenesis, 2) developing a system to monitor the need for countermeasure administration during or after flight, and 3) development of countermeasures that can be applied before, during and after flight. 2. Key Findings. We have optimized RNA isolation procedures so that we routinely obtain sufficient quantities of RNA that is of very high quality making it possible to conduct microarray analyses on our samples. We have also developed a normalization process for microarray data that deals with the problem of partially degraded materials such as that present in fecal material (Liu et al., 2005). Analysis of fecal samples collected over time indicated there were a variety of genes with differences in expression in response to diet, radiation treatment, stage of disease progression and their interactions. The most dramatic change was observed at the tumor end point in the rats consuming the corn oil and cellulose diets that had been exposed to radiation. A gene ontology analysis indicated a variety of genes affected by the diet X treatment interaction that were involved in carbohydrate and lipid metabolism and biosynthesis. We are in the process of determining the specific roles of these genes in colon carcinogenesis and thus, estimating their predictive power in a non-invasive screening program. 3. Impact of Findings on Project Goals. We have been successful in implementing the procedures necessary to perform RNA isolation from mucosa and fecal samples. RNA quality and quantity has been sufficient to perform the needed analyses. Therefore, we are on track in order to achieve the goals described in specific aim #1 and #2 of this project. 4. Proposed Research Plan for the Coming Year. The isolated fecal RNA will be used for microarray analyses. To prepare for this we will need to perform the appropriate reverse transcription reactions and label the RNA in preparation for array hybridizations. Results from the microarray analyses will be statistically analyzed and key genes identified so that PCR assays may be initiated in order to confirm the microarray results. Data generated last year will be analyzed and the results will be used to prepare publications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Information generated from our research project will easily extend to Earth medical research. Individuals exposed to radiation on Earth, either through their profession, or for diagnostic or therapeutic radiation procedures will benefit from the knowledge gained. Identification of gene expression patterns that are indicative of colon cancer development would make screening procedures less invasive, and thus hopefully improve early disease detection.
Task Progress:	This project aims to understand the changes in gene expression that occur in the colon after exposure to radiation, followed by exposure to a chemical carcinogen. One goal is to understand disease progression and to identify gene expression patterns that are indicative of the disease stage. Our second goal is to compare gene expression profiles of mucosal samples to samples of RNA from sloughed cells in order to develop a non-invasive procedure of detecting the changes. Lastly, we wanted to understand how diets containing either corn oil or fish oil as the lipid component and either cellulose or pectin as the fiber source affected the exposure to radiation and the carcinogen, and thus disease progression. Over the last year we completed all of the microarray analyses of the mucosal samples (n = 212) and the resulting data are currently undergoing initial statistical analyses. Secondly, we have completed the isolation of all the fecal polyA samples. We are currently finalizing our assays to establish the quantity of RNA isolated for each sample, which is the final step prior to starting microarray processing. Analyses of fecal samples collected at three times after carcinogen injection (7, 14 or 28 weeks) identified genes affected by diet, treatment and stage of disease, as well as their interactions. Rats consuming corn oil and cellulose and exposed to radiation had many gene clusters demonstrating different patterns of expression, especially at the 28-week sample, which is near the tumor end point. Gene ontology analysis of these significant genes yielded several genes involved in carbohydrate and lipid metabolism and biosynthesis. We are in the process of identifying the potential roles in colon carcinogenesis for these genes and identifying a set of genes that are most predictive of the tumor endpoint data at the various collection times.
Bibliography Type:	Description: (Last Updated: 08/22/2020)
Abstracts for Journals and Proceedings	Hong MY, Turner ND, Murphy ME, Carroll RJ, Bancroft LK, Davidson LA, Chapkin, Lupton JR. "Dietary fish oil down-regulates pro-inflammatory gene expression in colonocytes." Experimental Biology, San Francisco, CA, April 1-5 , 2006. FASEB J. 2006;20:A150. , Apr-2006
Abstracts for Journals and Proceedings	Lupton JR, Turner ND, Braby LA, Ford JR, Carroll RJ, Chapkin RS. "A combination of omega-3 fatty acids and a butyrate-producing fiber mitigates colon cancer development." 57th International Astronautical Federation Meeting Congress, Valencia, Spain, October 2-6, 2006. 57th International Astronautical Federation Meeting Congress, October 2006. , Oct-2006
Abstracts for Journals and Proceedings	Vanamala J, Glagolenko A, Carroll RJ, Murphy ME, Taddeo SS, Chapkin RS, Turner ND, Lupton JR. "Combination of fish oil and pectin suppressed beta catenin nuclear translocation, an important molecular event in colon carcinogenesis." AICR annual meeting, Washington, DC, July 13, 2006. AICR annual meeting, July 2006. , Jul-2006

Abstracts for Journals and Proceedings	Vanamala J, Glagolenko A, Carroll RJ, Murphy ME, Taddeo SS, Chapkin RS, Turner ND, Lupton JR. "Fish oil and pectin enhance apoptosis in irradiated rat colonocytes via suppression of PGE synthase-2 and Wnt pathway." Experimental Biology, San Francisco, CA, April 1-5 , 2006. FASEB J. 2006;20:A993. , Apr-2006
Abstracts for Journals and Proceedings	Zhou GD, Richardson M, Lupton JR, Turner ND, Chapkin, Donnelly KC. "Colonic cyclopurines induced by azoxymethane and irradiation are decreased by dietary fish oil." 97th Annual Meeting of the American Association for Cancer Research, Washington, DC, April 1-5, 2006. Proc Amer Assoc Cancer Res. 2006;47:abstract #1910. , May-2006
Articles in Peer-reviewed Journals	Baladandayuthapani V, Mallick BK, Hong MY, Lupton JR, Turner ND, Carroll RJ. "Bayesian hierarchical spatially correlated functional data analysis with application to colon carcinogenesis." Biometrics. In press, October 2006. , Oct-2006
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Awards	Lupton J. "Dr. Lupton was elected President elect of the American Society for Nutrition for 2006." Oct-2006
Awards	Lupton J. "Dr. Lupton was elected to membership in the International Astronautical Federation." Oct-2006
Awards	Lupton J. "Dr. Lupton was elected to serve on the Board of Directors for the International Life Sciences Institute." Oct-2006